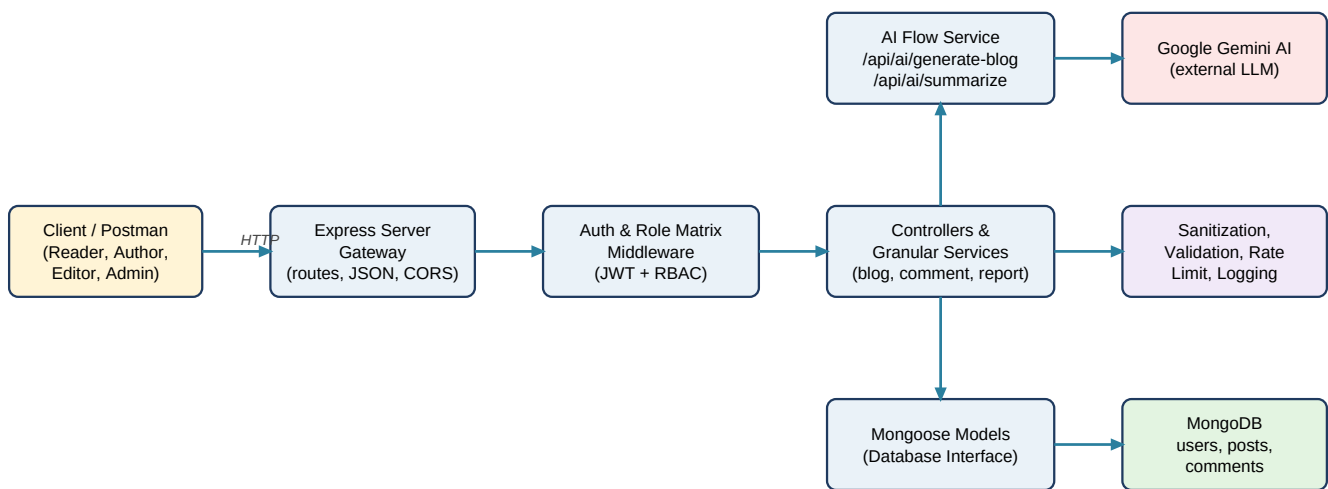


Technology Stack (Architecture & Stack)

Date	29/09/2026
Team ID	-
Project Name	BlogNest - AI-Powered Blog Management Backend
Maximum Marks	-

Technical Architecture:



Technical architecture of BlogNest

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Headless service; clients such as Postman, ThunderClient or any web or mobile frontend call the REST API	Postman / ThunderClient, any JavaScript frontend
2	Application Logic-1	Server, routing, controllers and blog lifecycle logic	Node.js, Express.js
3	Application Logic-2	Authentication and authorization	JWT, bcrypt.js, role-based middleware
4	Application Logic-3	AI blog generation and summarization	Google Gemini AI via /api/ai/generate-blog and /api/ai/summarize
5	Database	Stores users, posts, categories, comments and analytics as documents	MongoDB with Mongoose ODM
6	Cloud Database	Optional managed database hosting	MongoDB Atlas (optional)
7	File Storage	Media paths and metadata are stored; files kept on local filesystem or external storage	Local filesystem / cloud storage (optional)
8	External API	AI content generation and summarization	Google Gemini API

S.No	Component	Description	Technology
9	Machine Learning Model	Large language model for text generation and summarization	Google Gemini (LLM)
10	Infrastructure (Server / Cloud)	Local: Node.js v16+ and npm v8+ on Windows, macOS or Linux; 8 GB RAM (16 GB recommended). Cloud: any Node-capable host	Local server; cloud deployment optional

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source runtime, web framework, ODM and tooling	Node.js, Express.js, Mongoose
2	Security Implementations	Stateless auth, role checks, password hashing, sanitization, validation and rate limiting	JWT, bcrypt.js, RBAC, input sanitization (XSS, NoSQL injection), rate limiting
3	Scalable Architecture	Modular MVC layers; stateless server allows horizontal scaling	MVC, layered services
4	Availability	Multiple stateless server instances behind a load balancer; replicated database	Load balancer, MongoDB replica set (deployment option)
5	Performance	Indexed queries, lightweight routing, AI work offloaded to an external service	MongoDB indexes, Express, async processing